Supplementary File S7: November 1–5 Simulator Evaluation Evidence

November 5, 2026

Scope and evidence selection

This file documents the simulator evidence cited in the revised evaluation. It keeps the integrated pipeline, scene–protocol probes, five-seed policy comparison, process-window ablation, approval pilot, production-mode checks, and the negative live-Agent pilot separate because they use different test units and denominators. The raw CSVs, run manifests, reports, and the self-contained visual report are in `Supplementary-File-S7-data-2026-11-04.zip`. No physical-plant result is included.

Consolidation update (November 2–3, 2026)

The data package is organized into four folders. `01-paper-evidence/` is the frozen November 1 batch used for the six principal experiment blocks. `02-c3-pilot-negative/` contains a November 2 live-Agent multi-round pilot. The archive retains its original folder name; here it is referred to as the negative live-Agent pilot. After pipeline fixes, the best episode completed with zero governed writes while the process variable remained outside its band, and a second episode timed out without a write. The pilot exposed platform integration defects in task-artifact addressing, lead-queue draining, and goal completion without checking the process criterion. These defects are outside the positive claims in this paper. `03-system-validation-not-used/` contains other integration and demonstration reports that are not pooled with the controlled experiment blocks. `04-followup-scenario-pilot/` contains the November 3 single-run injection-twin pilot reported separately in Section VI; it is not a replication of the frozen batch.

A cast-film probe record from run `20261101121700-scenario-probes` was excluded. Its manifest declared four scenes but recorded only cast-film, and its simulator endpoint did not match the endpoint in the reproduction command. None of its 24 admissible requests, 24 structural probes, or 20 readbacks are included in the manuscript totals or the data package. The reported scene–protocol counts below use only run `20261101121221-scenario-probes`.

November 5 follow-up records

A corrected archive, `Supplementary-File-S7-followup-2026-11-05-r1.zip`, preserves the matrix CSV and manifest, the readback-mismatch record and an interpretation erratum, the two Agent outcome records and logs, the environment preflight, and the batch report. Its root manifest covers every other archive member. The four-scene matrix run `20261105060237-scenario-probes` used a recipe-window fraction of 0.8. All 110 collected requests outside those windows but inside structural limits returned `RECIPE_LIMIT_EXCEEDED`. Its other counts (220/220 accepted, 219/219 structural rejections, and 216/216 matching available readbacks) are separate from, and not added to, the November 1 batch. Input values and checks share configuration limits.

In run `20261105060514-readback-mismatch`, one simulated injection Modbus TCP node was subject to a calibration-offset injection: the command was 230, the driver readback was 251, and tolerance was 0.7. The outcome was `ok=false` and `unverified`; a separate read of the same simulator also returned 251. The 21-unit command-to-readback difference triggered the degraded receipt, while the separate read agreed with the driver's value; it was not independent hardware evidence. One baseline and one recovery returned `readback-verified`. The two November 5 Agent episodes were canceled after 371.63 and 372.24 seconds with zero rounds and zero writes; their recorded 12 minutes is a budget, not elapsed time. The batch report marks the Agent result unknown and the overall batch not passing.

The matrix manifest records commit `3f30a146` and harness hash `047818ab79d61b83`; the available runner and two scenario library files reproduce that hash. The services were already running, so the commit is not proof of their deployed versions or a complete configuration snapshot. The mismatch record has no runner hash. The

first November 5 follow-up archive omitted three artifacts named in its original run checksum list: the preflight `qualify.json` record and two Agent logs. The corrected `-r1` archive restores them and provides a new root manifest. The separate expanded November 5 archive retains a stale root manifest and should not be used as a complete package inventory. These records support selected simulator paths, not physical actuation or a process-efficacy estimate.

Representative systems

Table S7-I summarizes the operation boundary and evaluation scope of representative plant-facing agent systems cited in the manuscript. It does not rank their performance, and the reported evaluations are not directly comparable because the systems, tasks, and denominators differ.

Table 1: Representative plant-facing agent systems and the scope of their reported evaluations.

System	Reported operation or integration mechanism	Reported evaluation
Biskupovic et al. [1]	Agent-based service selection and online reselection using performance and computational-load metrics	Industrial cyber–physical system study; service adaptability and performance
Köhle et al. [2]	Module-specific agents, MCP-to-OPC-UA skills, MQTT coordination, and injected factory state	Nine simulated production challenges
Xavier et al. [3]	MCP adapters for industrial protocols with a common response envelope and mock-first testing	870 benchmark runs and 2,820 tool calls
Graul et al. [4]	AAS-grounded proposals, documented human authorization, and PLC-validated OPC UA Methods	Two laboratory proof-of-concept settings
Schall [5]	P&ID-, SCADA-, and SOP-grounded proposal verification with gated execution	Five simulated scenarios, 43 proposals, and 50 LLM runs
Lin and Chen [6]	Policy-governed sensor–MES–ERP digital thread	Manufacturing-record evaluation with operational metrics
Vyas et al. [7]	Multi-agent fault-recovery planning with Digital Process Plant Twin validation before actuation	Mixing and continuous stirred-tank reactor studies
PLCBench [8]	Hardware-in-the-loop PLC security benchmark with closed-loop process simulation	Security evaluation; complementary to operation-governance tests
This work	Scene-bound production context and named signal mappings; shared dispatch with route-specific write governance and separate command receipts and process-observation evidence	Two pipeline runs (77 assertions each), a four-scene probe subset, five-seed scripted missions, a three-repeat mock-driver ablation, one exploratory injection-twin pilot, and a negative live-Agent pilot

Integrated pipeline and scene–protocol probes

The integrated executions `20261101103641-9j8` and `20261101104849-xu0` used seed 42 and reported commit `da627d21`. Both manifests record that the platform and simulator were already running and that no Agent harness was active. Each run returned 77 passing assertions, zero warnings, and zero failures. Assertions are within-run checks, not independent trial observations.

Table 2: Valid collected scene–protocol probe outcomes.

Scene	Writable nodes	Writable nodes by protocol	Admissible accepted	Structural rejected	Readback matched
Injection	11	TCP 5; RTU 2; OPC UA 3; MQTT 1	44/44	44/44	44/44
Wastewater	7	TCP 1; RTU 3; OPC UA 2; MQTT 1	28/28	28/28	28/28
Annealing	7	TCP 3; RTU 2; OPC UA 1; MQTT 1	28/28	28/28	28/28
Biaxial line	30	TCP 20; RTU 1; OPC UA 8; HTTP 1	120/120	119/119	116/116

TCP and RTU in the table denote Modbus TCP and Modbus RTU. For each writable node, four tested values were expected to pass (three interior values and the upper boundary), followed by four structural-range probes expected to reject. One structural case in the biaxial-line set was not collected. Readback is counted only when

the driver returned an actuator value. Across this four-scene subset, the totals are 220/220 accepted admissible requests, 219/219 rejected structural probes, and 216/216 matching available readbacks. The full planned scene–protocol matrix remains incomplete.

The integrated film-line preset check is a separate pipeline result: five endpoint types were configured and nine selected negative probes were rejected (9/9), with no code edits. It reused the same runtime and is not an independent cold-start deployment. The pipeline run manifests and summaries are included in the data archive.

Five-seed target-band comparison

The no-write control ran for 60 s; its endpoint is the mean of the final five process-variable samples. The scripted setpoint policy used the `recipe_trial` batch route of the shared governed-write service with scenario-specific mission logic; its endpoint is the final value recorded after the scenario-specific steps, not a fixed-horizon measurement. Each seed appears in both conditions for injection mass, wastewater dissolved oxygen (DO), and annealing hardness. The 150.41–172.90 s range records the wall time of each three-scene parallel batch, including setup; it is not the per-scene time to target. The contrast is descriptive rather than a time-matched test. The seed-42 no-write runs began from residual state (injection 32.378 g and DO 2.894 mg/L, both inside their bands), whereas the scripted runs and seeds 43–46 began from the scenario blueprint. Counts are reported for 12 blueprint-matched pairs and for all 15 runs. We recomputed band membership from the raw `pv`, `target`, and `tol` fields. The runner's `attained` field combines condition-specific checks and is not compared across conditions.

Table 3: Raw endpoints with band membership recomputed from each process variable, target, and tolerance. The final column lists no-write/scripted outcomes: the no-write control was in band in 0/15 runs and the scripted policy in 15/15; for the 12 blueprint-matched pairs, the counts were 0/12 and 12/12.

Scene	Target band	Seed	No-write endpoint	Scripted endpoint	In band
Injection	32.5 ± 0.35 g	42	31.226	32.320	No / Yes
Injection	32.5 ± 0.35 g	43	31.272	32.180	No / Yes
Injection	32.5 ± 0.35 g	44	31.244	32.173	No / Yes
Injection	32.5 ± 0.35 g	45	31.272	32.190	No / Yes
Injection	32.5 ± 0.35 g	46	31.206	32.340	No / Yes
Wastewater DO	3.4 ± 0.6 mg/L	42	0.202	3.864	No / Yes
Wastewater DO	3.4 ± 0.6 mg/L	43	0.198	3.800	No / Yes
Wastewater DO	3.4 ± 0.6 mg/L	44	0.268	3.686	No / Yes
Wastewater DO	3.4 ± 0.6 mg/L	45	0.200	3.576	No / Yes
Wastewater DO	3.4 ± 0.6 mg/L	46	0.226	3.646	No / Yes
Annealing	95 ± 6 HV	42	134.66	97.74	No / Yes
Annealing	95 ± 6 HV	43	134.60	97.44	No / Yes
Annealing	95 ± 6 HV	44	134.66	97.80	No / Yes
Annealing	95 ± 6 HV	45	134.90	99.90	No / Yes
Annealing	95 ± 6 HV	46	134.36	98.20	No / Yes

The individual values above are copied from the ten per-seed CSV result files. Across the 12 blueprint-matched pairs, 0/12 no-write endpoints and 12/12 scripted-policy endpoints were in band; including the residual-state seed-42 control gives 0/15 and 15/15. Across the five scripted-policy seeds, the missions issued 12 writes in 17 rounds for injection, 56 writes in 59 rounds for wastewater, and 41 writes in 50 rounds for annealing. These are workload counts, not an efficiency comparison. Seeds are simulator initializations, not independent production lines. The scripted policy is not an active-LLM policy, and the no-write control is not an industrial-controller baseline.

Process-window guard ablation

The process-window study used seed 42, three repetitions in each arm, a direct DCW API call, a mock driver, and benchmark mode. The configuration hash was `3a57e722d4764b98`. Each repetition submitted six negative probes: four hard structural probes and two process-window probes; it also submitted three nominal legal requests and

three boundary requests. The additional 20 timed writes per repetition were used for latency summaries and were excluded from these denominators. No readback-mismatch stimulus was included.

Table 4: Three-repeat process-window ablation. The combined-removal arm removes the process-window and readback-mismatch checks; hard structural limits remain active in every arm.

Arm	Negative probes rejected	Window violations returning HTTP 200	Legal false blocks	Boundary expectations met
Full guard	18/18	0/6	0/9	9/9
No process-window guard	12/18	6/6	0/9	9/9
No-readback check	18/18	0/6	0/9	9/9
No process-window or readback guard	12/18	6/6	0/9	9/9

The four hard structural probe types were blocked in every arm (12/12 per arm). With the process-window guard active, all six targeted window violations returned HTTP 400; removing that guard yielded HTTP 200 for all six. The archive lacks response bodies and per-request driver-call logs; status alone does not establish invocation of the mock driver. The no-readback arm does not test readback-mismatch rejection because that stimulus was absent. Benchmark mode relaxes normal timing and hold-window controls, so these results concern the tested guard path only.

Approval, production-mode, and authorization checks

The approval pilot used one injection scene and a scripted decision fixture. The full CSV and timing records are in the data archive.

Table 5: Four calls in the approval-route pilot. Auto rows are not clean matched pairs with the manual rows; no operator made a decision.

Mode	Runner stimulus	Decision	Before	After	Delay (ms)
Manual	legal-approved	approved	202	204.8	10 593
Manual	legal-rejected	rejected	216	216	544
Auto	label: manual arm rejected	n/a	202	204.8	10 071
Auto	label: manual arm approved	n/a	216	218.8	10 073

The manual reject did not change the simulated control value. The auto-mode label/value mismatch means the four calls do not form a clean paired approval-on/off experiment. This pilot demonstrates route handling only; it is not evidence of operator judgment or an isolated approval-mechanism effect.

In the separate production-mode guard run, the baseline value 190 was accepted. One example each was rejected for interval, step, node-range, recipe-window, and hold-window constraints. The exact API rejection codes are preserved in the source report and CSV. These one-case checks confirm expected branches for their configured inputs; they are not estimates of general rejection rates. A separate three-case authorization probe rejected an unbound write and a cross-node write while accepting one bound control write.

Follow-up exploratory injection scenario pilot

The run was recorded at 2026-11-03 07:43:08 UTC under plan identifier `pexp-mw0dmt4v-na04` for line `ln-00e0726d`. The isolated platform used a PLC-based simulated twin; the run manifest sets `realPlant=false`. An acceptance script executed the workflow. This pilot is separate from the frozen November 1 experiment batch, is not a replication of the five-seed comparison, and does not represent an active-Agent or physical-plant run. The acceptance report identifies a v0.9.4 starting point with same-PR fixes and a resulting v0.9.5 version. The individual run record contains no platform commit, simulator snapshot/configuration hash, or runner-source hash, so this pilot has less complete provenance than the frozen batch and is not a hermetic source snapshot.

At setpoint 50 bar, the structured record gives a baseline product-weight observation of 31.503 g, below the 32.00–32.30 g band. The accompanying report gives 31.508 g; the discrepancy has not been resolved. The experiment plan probed setpoints 50, 55, and 60 bar. The stored ledger fields are:

Table 6: Aggregate values and classifications saved by the single-run experiment ledger.

Setpoint contrast	Stored effect (g)	Stored MDE (g)	Stored verdict
50 to 50	0.00452	0.02214	`no_resolvable_effect`
50 to 55	0.26638	0.02007	`underpowered`
55 to 60	0.25173	0.02999	`identifiable_effect`

MDE denotes minimum detectable effect. These are the aggregate fields stored by the acceptance script; the archive contains neither the underlying sample series nor the rule used to calculate the MDE and assign the verdicts. The reported gain of 0.05035 g/bar is the last stored effect divided by the 5-bar step (0.2517336/5); it is a source-reported ratio, not an independently verified or replicated process model. After the design steps, the governed write changed the setpoint from 60 to 63.7; the returned device readback was 63.7. Following a 60-s wait, the runner averaged the three latest 15-s DAQ buckets and measured 32.224 g, within the configured band. The reported 15/15 count refers to script assertions over this single execution, not to 15 independent trials.

Table 7: Summary of the single follow-up injection-twin execution.

Stage	Recorded result	Interpretation limit
Baseline	31.503 g at setpoint 50 bar	One simulated-twin observation
Design of experiments	Levels 50, 55, and 60 bar; one ledger verdict marked identifiable	Sample series and verdict rule are not in the archive
Gain and final write	Reported ratio of 0.05035 g/bar; write from 60 to 63.7 with matching readback	Single-run ratio; not a replicated process estimate
Process observation	32.224 g after the specified wait and sampling window; target band 32.00–32.30 g	One in-band endpoint, no replication or physical-plant result

The source report, machine-readable result, and execution log are in `04-followup-scenario-pilot/`. The local checksum manifest identifies the included files. Later platform acceptance suites use other fixtures and assertion outcomes and are not pooled with this pilot or the six principal experiment blocks.

Reproduction and provenance

The platform manifests record source commit `da627d21`. The experiment runners were uncommitted at execution time and were later committed as `40533741`; the recorded commit therefore identifies the platform source, not a hermetic tree. The raw source records in the archive preserve runner outputs and exact run identifiers. The two raw condition keys below are retained because the runner requires them: `C0` denotes the no-write control, and `C1` denotes the scripted setpoint policy. Representative commands recorded by the runners are:

```
node bench/pipeline.mjs -profile integrated -seed 42 -cl-seeds 3

node bench/experiment/scenario-protocol-probes.mjs -scenarios injection,wwtp,anneal,biax -seed 42

node bench/experiment/optimization-control.mjs -condition C1 -scenarios injection,wwtp,anneal -seed 42

node bench/experiment/optimization-control.mjs -condition C0 -scenarios injection,wwtp,anneal -seed 42
-duration 60

node bench/e1-lite.mjs -seed 42 -repeats 3

node bench/experiment/approval-gate.mjs -scenario injection -base http://127.0.0.1:3005

node bench/experiment/production-guards.mjs -base http://127.0.0.1:3010

node bench/experiment/authorization-probe.mjs -scenario injection -base http://127.0.0.1:3005
```

The optimization commands were run for seeds 42–46 in both conditions. Re-running them creates new result identifiers and can produce different simulator trajectories if the environment or service state changes. The integrated pipeline reported a commit and harness hash, but the active service source tree was not captured as a hermetic snapshot. The archive includes the run manifests, selected reports, raw CSVs, a machine-readable validation summary, checksum manifest, and self-contained HTML dashboard; it does not include the excluded cast-film run.

References

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